Dhyan Patel

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Objective

Biomedical Engineering senior graduating December 2025 with a concentration in Bioinformatics. Seeking full-time roles in biomedical engineering, including medical devices, biotechnology, healthcare technology, and bioinformatics. Experienced in **engineering design, process optimization, quality assurance, and data analysis**, with a focus on advancing innovation and improving healthcare outcomes.

Education

University of Illinois at Chicago — Chicago, Illinois

Bachelor of Biomedical Engineering (Concentration in Bio Informatics) — Aug 2021 - **Dec 2025**

Passion Project Heated Bed for Mice During Ophthalmic Experimentation

- Designed and prototyped a flexible, waterproof heated bed with a resistive heating element to maintain anesthetized rodents' body temperature within 36–38 °C.
- Developed a cutting-edge, flexible, and waterproof heating system featuring a PID-controlled thermal regulation mechanism, ensuring precise and consistent heat distribution.
- Conducted in-depth patent research, design optimization, and prototyping, achieving a scalable solution compliant with **ISO 13485:2018** and **IEC 60601-2-35** medical standards.

Work Experience

Manufacturing Engineering Intern | India Flex Engineering | Ahmedabad, Gujarat — **May 2022 - Aug 2022**

- Supported the design and manufacture of **metallic/non-metallic expansion joints and pressure vessels**, creating/maintaining process documentation and tooling setups in accordance with **ISO 9001:2015**.
- Assisted in **quality assurance procedures**, ensuring products met stringent specifications and customer expectations, which improved product reliability and performance through collaboration with cross-functional teams
- Analyzed and optimized manufacturing processes, contributing to efficient production workflows and advanced material selection for expansion joints and pressure vessels, which supported India Flex Engineering's focus on innovative engineering solutions

Computer specialist | University of Illinois Chicago | Chicago, IL — **Jun 2023 - Present**

- Diagnose and resolve complex hardware, OS, and application issues for faculty/staff; perform component-level repairs, imaging, and preventative maintenance to maximize endpoint uptime.
- Install, configure, and document systems, drivers, and peripherals; manage user access and device enrollment while ensuring secure, reliable wired/wireless connectivity to campus networks.
- Streamline support workflows by documenting incidents/SOPs and coaching users on best practices, improving service consistency and reducing repeat issues.

Learning Assistant | University of Illinois Chicago | Chicago, IL — **Aug 2022 - Present**

- Facilitate active-learning workshops and problem-solving sessions, improving student comprehension and course outcomes by **15%** through collaborative instruction and formative feedback.
- Support students on assigned tasks and concepts; develop clarifying examples, checkpoints, and study guides to strengthen application of core material and reduce rework.
- Partner with instructors to refine course flow, rubrics, and elevating overall instructional efficiency and performance.

Academic Projects

Wearable ECG & Respiratory Rate Monitor

- Architected a wearable using the SparkFun AD8232 ECG module and an FSR on an elastic chest band; selected Lead I/II electrode placement and shielding/grounding to reduce motion and mains artefacts.
- Implemented real-time signal conditioning: band-pass (≈0.5–40 Hz), 60 Hz notch, baseline-wander removal and moving-average smoothing; streamed samples from Arduino for live processing.
- Derived breathing rate by low-pass filtering the FSR signal and using peak/zero-crossing detection; added calibration routine to map FSR units to respiratory cycles. Built a Processing/MATLAB GUI with all the signals data

SpO_2 + Heart-Rate Wearable

- Prototyped a low-power pulse oximeter using the MAX30102 (LED current/ADC/sample-rate config); implemented moving-average filtering and peak detection to derive HR and SpO_2.
- Streamed vitals via BLE to a cross-platform dashboard; logged timestamped CSV and computed basic HRV metrics (e.g., RMSSD, SDNN) for analysis.
- Modeled the wrist enclosure and sensor mount in AutoCAD (light-blocking shroud, strap slots, strain-relief); exported STL for 3D printing and iterated fit to minimize optical leakage and motion artefacts.

Skills

- **Design & Engineering Tools:** CAD Software, Solid Works, Medical Device Design, Circuit Design and Analysis**,**PCB design
- **Programming:** Python, MATLAB, LabVIEW, Data Modeling and Statistical Analysis
- **Bio-Med:** Bio-signal Processing, Biomedical Instrumentation, ISO Standards, Biomaterials, FDA Regulations, Bioinformatics Tools, Biomedical Thermodynamics